Area of Triangles, Parallelograms and Trapeziums

Name _____ Date _____ No calculator Version 1 Scan to practise this topic online

KEY WORDS area *(horahanga)* · base *(pūtake)* · perpendicular height *(teitei hāngai)* · triangle *(tapatoru)*

A Rectangles and triangles ●○○

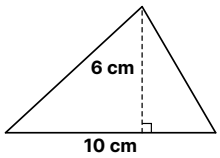
Remember: Area of a rectangle $= l \times w$. Area of a triangle $= \frac{1}{2} \times b \times h$, where the height is at right angles to the base.

1. Find the area of each shape.

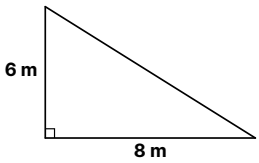
a)



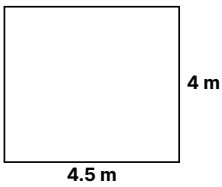
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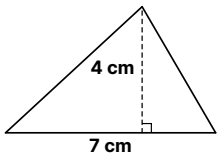
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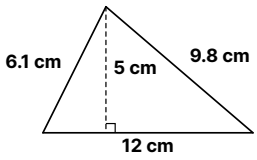
d)



e)



f)



2. Find the area of:

a) A square with sides of 7 cm

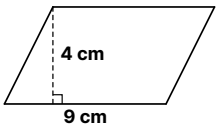
b) A triangle with base 2.5 m and height 4 m

B Parallelograms ●●○

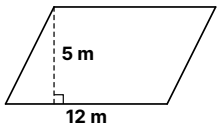
Watch out: Area of a parallelogram $= b \times h$. Use the height that is at right angles to the base, not the length of the sloping side.

3. Find the area of each parallelogram.

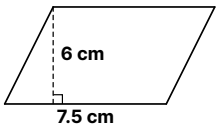
a)



b)



c)



4. Find the area of:

a) A parallelogram with base 15 mm and height 8 mm

b) A parallelogram with base 2.4 m and height 5 m

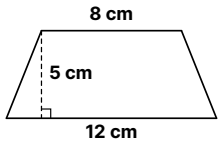
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C Trapeziums ●●○

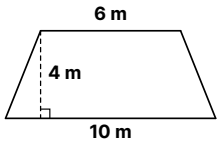
> **Remember:** Area of a trapezium $= \frac{1}{2}(a + b) \times h$: add the two parallel sides, halve, then multiply by the height between them.

5. Find the area of each trapezium.

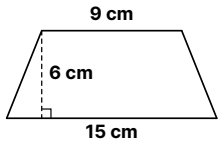
a)



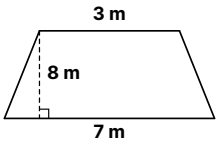
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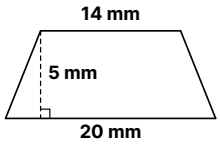
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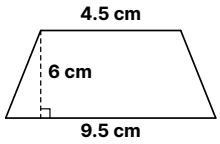
d)



e)



f)



6. Find the area of:

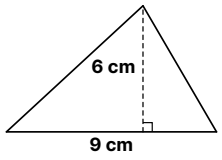
a) A trapezium with parallel sides of 14 cm and 10 cm, and a height of 7 cm

b) A trapezium with parallel sides of 2.5 m and 1.5 m, and a height of 3 m

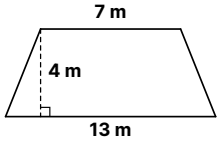
D Choosing the formula ●●●

7. Decide which formula each shape needs, then find its area.

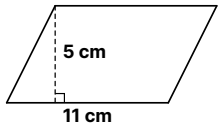
a)



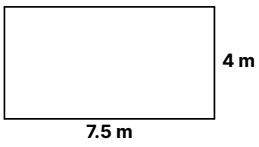
b)



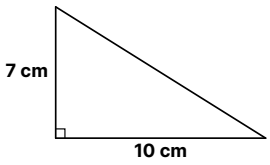
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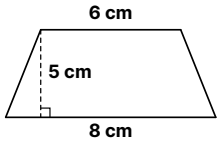
d)



e)



f)



E Problems ●●●

8. A garden is a trapezium. Its parallel sides are 18 m and 12 m long, and they are 10 m apart. Find the area of the garden.

9. A triangular sail has a base of 6 m and a height of 4 m. Sail cloth costs $2.50 for each square metre. How much does the cloth for the sail cost?

10. A field is a parallelogram with a base of 40 m and a height of 25 m. A farmer wants 150 m² for each of her sheep. What is the largest number of sheep she can keep in the field?

★ Extension

11. A triangle has an area of 30 cm² and a base of 12 cm. Find its height.

12. A trapezium has an area of 36 cm². Its parallel sides are 10 cm and 8 cm long. Find its height.

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Area of Triangles, Parallelograms and Trapeziums

Name _____ Date _____ No calculator Version 2 Scan to practise this topic online

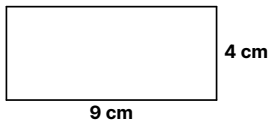
KEY WORDS area *(horahanga)* · base *(pūtake)* · perpendicular height *(teitei hāngai)* · triangle *(tapatoru)*

A Rectangles and triangles ●○○

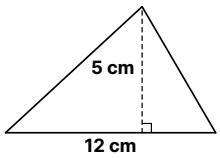
Remember: Area of a rectangle $= l \times w$. Area of a triangle $= \frac{1}{2} \times b \times h$, where the height is at right angles to the base.

1. Find the area of each shape.

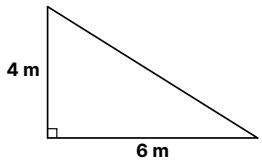
a)



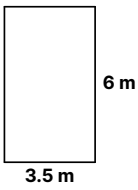
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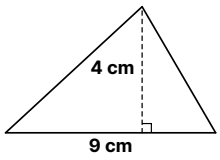
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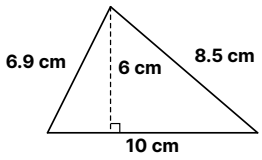
d)



e)



f)



2. Find the area of:

a) A square with sides of 9 cm

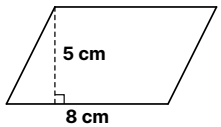
b) A triangle with base 3.5 m and height 4 m

B Parallelograms ●●○

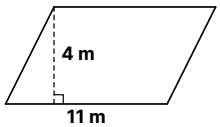
Watch out: Area of a parallelogram $= b \times h$. Use the height that is at right angles to the base, not the length of the sloping side.

3. Find the area of each parallelogram.

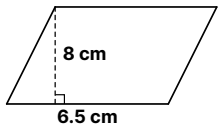
a)



b)



c)



4. Find the area of:

a) A parallelogram with base 14 mm and height 9 mm

b) A parallelogram with base 3.2 m and height 5 m

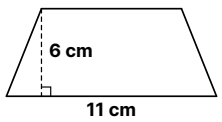
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C Trapeziums ●●○

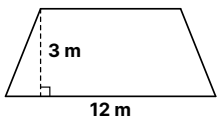
> **Remember:** Area of a trapezium $= \frac{1}{2}(a + b) \times h$: add the two parallel sides, halve, then multiply by the height between them.

5. Find the area of each trapezium.

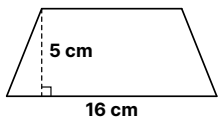
a)



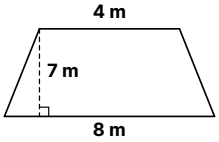
b)



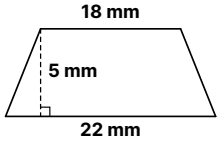
c)



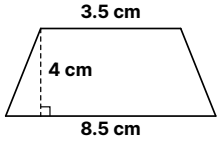
d)



e)



f)



6. Find the area of:

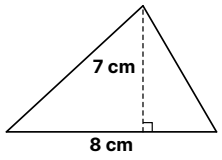
a) A trapezium with parallel sides of 13 cm and 9 cm, and a height of 8 cm

b) A trapezium with parallel sides of 3.5 m and 2.5 m, and a height of 2 m

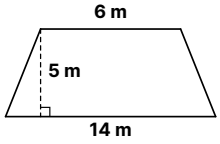
D Choosing the formula ●●●

7. Decide which formula each shape needs, then find its area.

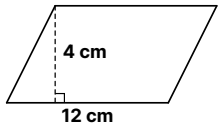
a)



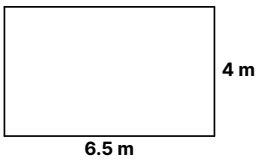
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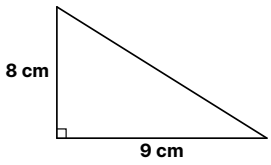
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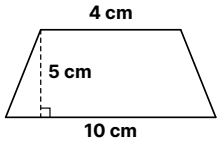
d)



e)



f)



E Problems ●●●

8. A garden is a trapezium. Its parallel sides are 20 m and 14 m long, and they are 8 m apart. Find the area of the garden.

9. A triangular sail has a base of 8 m and a height of 3 m. Sail cloth costs $2.50 for each square metre. How much does the cloth for the sail cost?

10. A field is a parallelogram with a base of 50 m and a height of 30 m. A farmer wants 200 m² for each of her sheep. What is the largest number of sheep she can keep in the field?

★ Extension

11. A triangle has an area of 42 cm² and a base of 14 cm. Find its height.

12. A trapezium has an area of 40 cm². Its parallel sides are 12 cm and 8 cm long. Find its height.

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